

"From Messy Data to Meaningful Data"

DSA0503-Query Processing For data sciences

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WHAT IS DIRTY DATA?

- **Duplicate Data:** The same information is stored multiple times.
- **Missing Data:** Important values are not available.
- **Incorrect Data:** Data contains wrong information.
- **Outdated Data:** Old information is no longer useful.
- **Inconsistent Data:** Data follows different formats.

WHY IS DIRTY DATA EXPENSIVE?

- **Wasted Time:** Employees spend time fixing incorrect data.
- **Lost Money:** Poor data can cause unnecessary expenses.
- **Bad Decisions:** Wrong data leads to wrong business decisions.
- **Lost Customers:** Incorrect information can reduce customer trust.
- **Poor Reports:** Dirty data produces inaccurate reports.

SOURCES OF DIRTY DATA

- **Manual Entry:** Humans can make mistakes while entering data.
- **System Errors:** Software problems can create incorrect records.
- **Multiple Systems:** Different databases may contain different information.
- **Old Records:** Stored information may become outdated.
- **Duplicate Entry:** The same data may be entered repeatedly.

FINDING DIRTY DATA USING QUERIES

- **Missing Values:** Queries can identify records with empty fields.
- **Duplicates:** Queries can find repeated records.
- **Wrong Values:** Queries can detect values outside expected ranges.
- **Filtering:** Queries can isolate problematic records.

COST OF DUPLICATE DATA

- **Duplicate Customers:** One customer may appear as multiple customers.
- **Wasted Resources:** Extra records increase storage and processing costs.
- **Wrong Analysis:** Duplicate data gives incorrect statistics.
- **Poor Marketing:** Customers may receive repeated messages.

BUSINESS IMPACT

- **Incorrect Data:** Leads to poor business decisions.
- **Missing Data:** Reduces the quality of analysis.
- **Duplicate Data:** Increases operational costs.
- **Outdated Data:** Can result in lost business opportunities.
- **Inconsistent Data:** Creates unreliable reports.

REDUCING THE COST

- **Clean Data Regularly:** Remove incorrect and outdated records.
- **Validate Data:** Check information before storing it.
- **Remove Duplicates:** Keep only unique records.
- **Monitor Quality:** Continuously check database accuracy.
- **Use Queries:** Quickly identify and analyze data problems.

SUMMARY

- **Dirty Data Costs Money:** Poor-quality data creates hidden business costs.
- **Queries Find Problems:** Query processing helps identify dirty data.
- **Clean Data Matters:** Accurate data produces reliable results.
- **Better Data, Better Decisions:** Quality data improves business performance.

GEO TAG

